

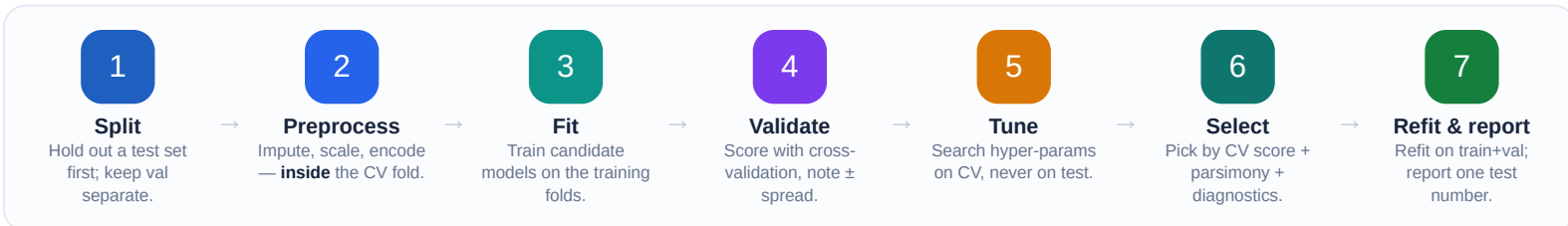
Build the Best Model: Workflow, Overfitting & Validation

The end-to-end path from raw data to a model you can trust — and the three numbers that tell you if it learned the signal or memorized the noise.

THE ONE RULE

The only score that estimates performance on **brand-new data** is **cross-validated**. Training and holdout are diagnostics — the CV mean $\pm$ spread is the truth.

THE END-TO-END MODELLING PIPELINE



🔒 Splitting & leakage guard

- ▶ **Train / Validation / Test** $\approx 60 / 20 / 20$. Touch test **once**, at the very end.
- ▶ **Stratify** on the label for classification so each split keeps class balance.
- ▶ **GroupKFold** when rows share a source (same slab/patient/user) — split by the group.
- ▶ **TimeSeriesSplit** for temporal data — train on the past, test on the future only.
- ▶ Fit scalers, imputers, encoders, and feature selection **inside** each fold (use a Pipeline).

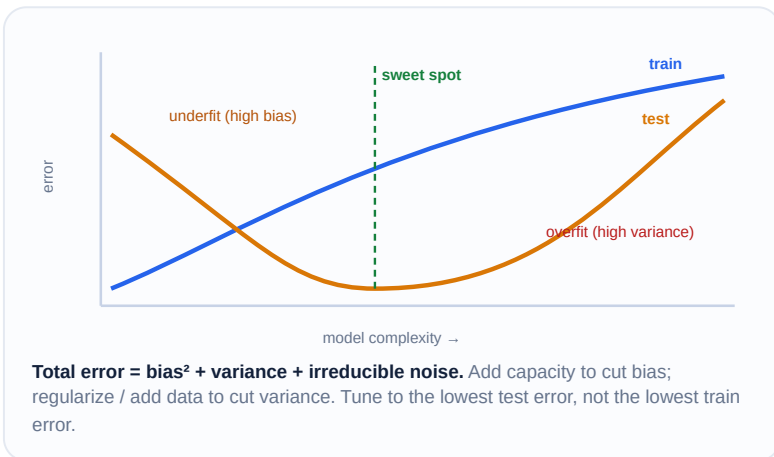
Leakage sources: target leakage (a feature encodes the answer), train $\rightarrow$ test contamination (preprocessing fit on all data), and temporal leakage (future info in past rows).

🔑 The three R^2 — read them together

SCORE	WHAT IT IS	TRUST
Training R^2	Fit on data the model has already seen.	Diagnostic only
Holdout R^2	One untouched 20% split.	Quick check
Cross-val R^2	Averaged over k held-out folds, with $\pm$ spread.	The headline

Overfit: high train, big drop to CV, wide $\pm$. **Underfit:** low everywhere. **Healthy:** all close & high, tight $\pm$.

BIAS-VARIANCE & THE COMPLEXITY SWEET SPOT



📈 Learning & validation curves

- ▶ **Learning curve** (score vs training-set size): gap that stays wide $\rightarrow$ variance, get more data / regularize. Both curves low & converged $\rightarrow$ bias, add capacity / features.
- ▶ **Validation curve** (score vs one hyper-parameter): find where CV peaks before it turns down.
- ▶ **Early stopping:** halt boosting / neural-net training when the validation score stops improving.

CROSS-VALIDATION SCHEMES — PICK THE ONE THAT MATCHES YOUR DATA

SCHEME	HOW IT WORKS	USE WHEN
k-Fold typically k = 5 or 10	Rotate each fold as the test set once; average the k scores.	Default for i.i.d. tabular data.
Stratified k-Fold	k-Fold that preserves class proportions in every fold.	Classification, especially imbalanced.
GroupKFold	No group appears in both train and test.	Repeated measures / clustered rows.
TimeSeriesSplit	Expanding window; always predict forward in time.	Any temporal / forecasting problem.
Repeated k-Fold	Run k-Fold several times with new shuffles; average.	Small data — tighten the estimate.
Nested CV outer + inner loops	Inner loop tunes hyper-params; outer loop measures the tuned model.	Honest score and tuning at once.

HYPER-PARAMETER SEARCH & CAPACITY CONTROL

🔧 Tuning strategies

- ▶ **Grid search** — exhaustive over a small, sensible grid.
- ▶ **Random search** — sample the space; wins when few params matter.
- ▶ **Bayesian optimization** (Optuna / Hyperopt) — model the score surface, spend trials where they help.

🔴 Regularization (shrink to fight variance)

Ridge (L2): $\min \sum (y - \hat{y})^2 + \lambda \sum \beta_i^2$ — shrinks coefficients smoothly.

Lasso (L1): $\min \sum (y - \hat{y})^2 + \lambda \sum |\beta_i|$ — drives some β to 0 (feature selection).

Elastic Net: mixes L1 + L2 ($\lambda[\alpha|\beta|_1 + (1-\alpha)|\beta|^2]$).

► **Successive halving** — start many configs cheaply, keep promoting the best.

Always tune against the **CV score**; keep the test set sealed.

Trees / NN: limit `max_depth` , `min_samples_leaf` , learning rate, subsample, dropout, and use early stopping.

★ TAKEAWAY

Split honestly, validate with cross-validation, tune on the CV score, then **refit and report one sealed test number**.
Fit and generalization are read together — never a single score.

UpSkill Sprint Consulting
upskillsprint.com

MODEL TUNING & EVALUATION · POSTER 2

NOTATION

`yi` actual · `ŷi` predicted · `ȳ` mean ·
`n` rows · `p` predictors.

Regression Metrics: Meaning, Formula & Acceptance Range

Every number the target is continuous — what it measures, its equation, and the range that counts as good. Match the metric to the cost of the error and the target's scale.

ERROR METRICS — SIZE OF THE MISS

METRIC	FORMULA	MEANING & WHEN TO USE	TARGET
MAE Mean Absolute Error	$(1/n) \sum y_i - \hat{y}_i $	Average miss in the target's own units; robust to outliers. Report to non-specialists.	lower; 0 = perfect
MSE Mean Squared Error	$(1/n) \sum (y_i - \hat{y}_i)^2$	Punishes big misses quadratically; the quantity most models optimize.	lower
RMSE Root Mean Squared Error	$\sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$	MSE back in target units; sensitive to large errors. Compare vs σ of y.	lower; < $\sigma(y)$
RMSLE Root Mean Sq. Log Error	$\sqrt{[(1/n) \sum (\ln(1+\hat{y}) - \ln(1+y))^2]}$	Penalizes relative error; good for skewed / multiplicative targets, under-prediction.	lower
MAPE Mean Abs. % Error	$(100/n) \sum (y_i - \hat{y}_i)/y_i $	Scale-free % error; fails if y ≈ 0. Common in forecasting.	<10% good
sMAPE symmetric MAPE	$(100/n) \sum 2 \hat{y}-y / (y + \hat{y})$	Bounded, symmetric version; handles small y better.	lower
MedAE Median Absolute Error	$\text{median}(y_i - \hat{y}_i)$	Fully outlier-robust typical miss.	lower
Huber Huber loss (δ)	$\frac{1}{2}a^2$ if $ a \leq \delta$; $\delta(a - \frac{1}{2}\delta)$ else	Quadratic near 0, linear in the tails — MSE accuracy with MAE robustness.	lower

GOODNESS-OF-FIT — SHARE OF VARIANCE EXPLAINED

METRIC	FORMULA	MEANING	RANGE
R² Coefficient of Determination	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Fraction of variance explained. Rises with any added predictor — never judge alone.	($-\infty$, 1]
Adjusted R²	$1 - \frac{(1 - R^2) \cdot (n-1)}{(n-p-1)}$	R ² penalized for predictor count; use to compare models of different size.	≤ R²
Explained Var.	$1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)}$	Like R ² but ignores systematic bias in the residuals.	($-\infty$, 1]

Acceptance is field-dependent: physical/engineering fits often expect R² > 0.9; noisy human-behaviour or social data may be strong at R² ≈ 0.3. Always compare against a baseline model.

MODEL SELECTION & PARSIMONY — FIT VS COMPLEXITY

CRITERION	FORMULA	RULE
AIC Akaike Information Criterion	$2k - 2 \ln(\hat{L})$	Lower is better. k = #parameters, $\hat{L}$ = max likelihood. $\Delta\text{AIC} > 10$ = decisive.
BIC Bayesian Information Criterion	$k \cdot \ln(n) - 2 \ln(\hat{L})$	Heavier complexity penalty than AIC → favours simpler models on big n.
Mallows' Cp	$\text{RSS}_p / \sigma^2 - n + 2p$	Good subset when $C_p \approx p$. RSS = residual sum of squares.
F-statistic	$(\text{explained}/p) / (\text{residual}/(n-p-1))$	Tests whether the model beats an intercept-only fit (overall significance).
Log-likelihood $\ln(\hat{L})$	$\sum \ln f(y_i \text{model})$	Higher = better fit; the raw input to AIC / BIC.

RESIDUAL DIAGNOSTICS — IS THE FIT EVEN TRUSTWORTHY?

CHECK	FORMULA / TEST	FLAG & ACTION
VIF Variance Inflation Factor	$\text{VIF}_j = 1 / (1 - R_j^2)$	> 5–10 → multicollinearity; drop/combine predictors or use ridge.

Condition number	$\sqrt{(\lambda_{\max} / \lambda_{\min})}$	$> 30 \rightarrow$ unstable coefficients; center/scale, remove redundancy.
Durbin–Watson	$\Sigma(e_t - e_{t-1})^2 / \Sigma e_t^2$	Range 0–4; ≈ 2 ideal. <1.5 or $>2.5 \rightarrow$ autocorrelation (order/robust SE).
Breusch–Pagan	χ^2 test on residual variance	$p < 0.05 \rightarrow$ heteroscedasticity; use robust SE or transform y.
Cook's distance	$D_i = (r_i^2/p) \cdot h_i / (1-h_i)$	$D_i > 4/n \rightarrow$ influential point; inspect, don't blindly delete.
Jarque–Bera normality of residuals	$(n/6)(S^2 + \frac{1}{4}(K-3)^2)$	Over-rejects on big n — judge by skew ≈ 0 , kurtosis ≈ 3 , and Q–Q plot.

★ TAKEAWAY

Report an **error metric in real units** (RMSE/MAE) with **R² / adjusted R²**, choose between models on **AIC/BIC**, and confirm the residual diagnostics before you believe a single coefficient.

UpSkill Sprint Consulting
upskillsprint.com

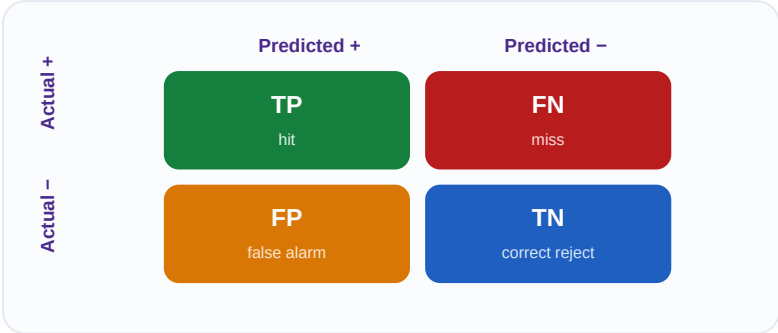
MODEL TUNING & EVALUATION · POSTER 3

COUNTS
TP true positive · FP false positive (false alarm) · FN false negative (miss) · TN true negative.

Classification Metrics & Tuning — with Equations

Everything the target is a class. Accuracy lies on imbalance — pick the metric that matches the mistake that actually costs you, then tune the threshold to it.

THE CONFUSION MATRIX — EVERYTHING IS BUILT FROM FOUR COUNTS



- 🚨 Which error costs more?
- ▶ Miss is dangerous (disease, defect, fraud) → maximize **Recall**.
 - ▶ False alarm is costly (spam block, audit) → maximize **Precision**.
 - ▶ Both matter → **F1 / Fβ**; ranking quality → **ROC-AUC / PR-AUC**.
 - ▶ Imbalanced → never accuracy; use **PR-AUC, MCC, balanced accuracy**.

THRESHOLD METRICS — FROM THE FOUR COUNTS

METRIC	FORMULA	MEANS
Accuracy	$(TP+TN) / (TP+TN+FP+FN)$	Share correct. Misleads badly on imbalance.
Precision Positive Predictive Value	$TP / (TP + FP)$	Of flagged, how many were real.
Recall Sensitivity · TPR	$TP / (TP + FN)$	Of real positives, how many caught.
Specificity TNR	$TN / (TN + FP)$	Of real negatives, how many cleared.
FPR False Positive Rate	$FP / (FP + TN) = 1 - \text{Specificity}$	False-alarm rate (x-axis of ROC).
F1-score	$2 \cdot (P \cdot R) / (P + R)$	Harmonic mean of precision & recall.
Fβ-score	$(1+\beta^2) \cdot P \cdot R / (\beta^2 \cdot P + R)$	$\beta > 1$ favours recall; $\beta < 1$ favours precision.
Balanced Acc.	$(\text{Sensitivity} + \text{Specificity}) / 2$	Accuracy corrected for imbalance.
Jaccard IoU	$TP / (TP + FP + FN)$	Overlap of predicted & actual positives.

PROBABILITY METRICS — SCORE THE RANKING & CALIBRATION

METRIC	FORMULA / DEFINITION	RANGE · TARGET
ROC-AUC Area Under ROC Curve	$P(\text{score}(\text{pos}) > \text{score}(\text{neg}))$	0.5 chance · 0.7 fair · 0.8 good · 0.9+ excellent
PR-AUC Avg. Precision	$\Sigma (R_n - R_{n-1}) \cdot P_n$	Better than ROC on rare positives; baseline = prevalence.
Log loss Binary cross-entropy	$-(1/n) \Sigma [y \ln p + (1-y) \ln(1-p)]$	lower; 0 = perfect ; punishes confident errors.
Brier score	$(1/n) \Sigma (p_i - y_i)^2$	lower ; mean-squared probability error (calibration).
MCC Matthews Corr. Coef.	$(TP \cdot TN - FP \cdot FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$	[-1, 1] ; balanced even on skew.

Cohen's κ
kappa

$$(p_o - p_e) / (1 - p_e)$$

Agreement above chance; >0.6 substantial.

 Threshold & curves

Default 0.5 is rarely optimal. Move the cut to trade precision ↔ recall; pick it by cost or by **Youden's J = TPR – FPR**. ROC plots TPR vs FPR; PR plots P vs R (use PR when positives are rare).

 Imbalance

- ▶ Class weights / `class_weight`
- ▶ Resample: **SMOTE**, under-sample
- ▶ Threshold moving
- ▶ Judge on PR-AUC, MCC

 Calibration & multiclass

Calibrate probabilities (Platt / isotonic); check a reliability diagram + Brier. **Multiclass avg:** macro (unweighted mean), micro (pool counts), weighted (by support).

★ TAKEAWAY

Read the **confusion matrix**, choose the metric that matches the costly error, tune the **threshold** (not just the model), and on imbalance trust **PR-AUC / MCC** over accuracy.